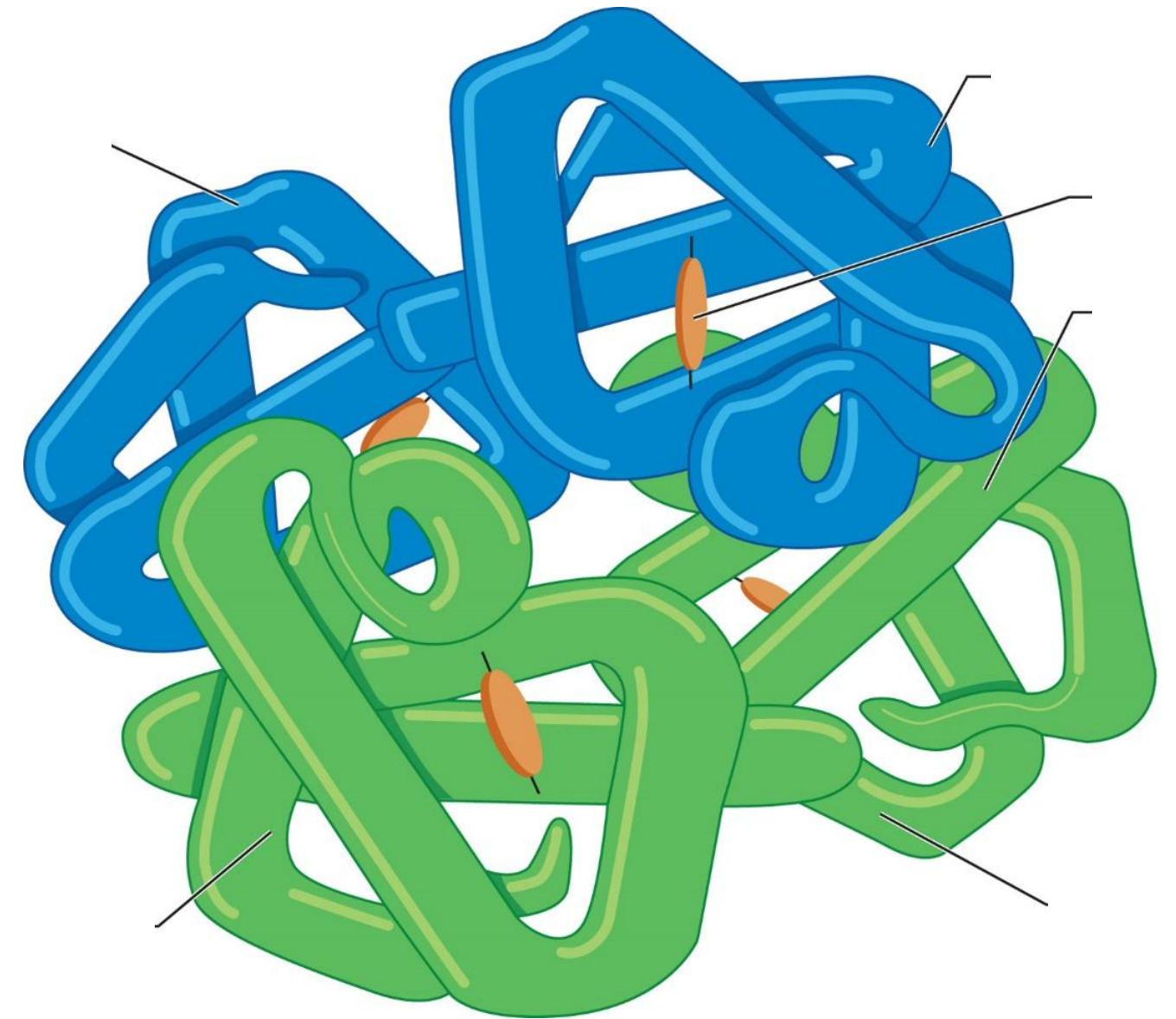


Hemoglobin

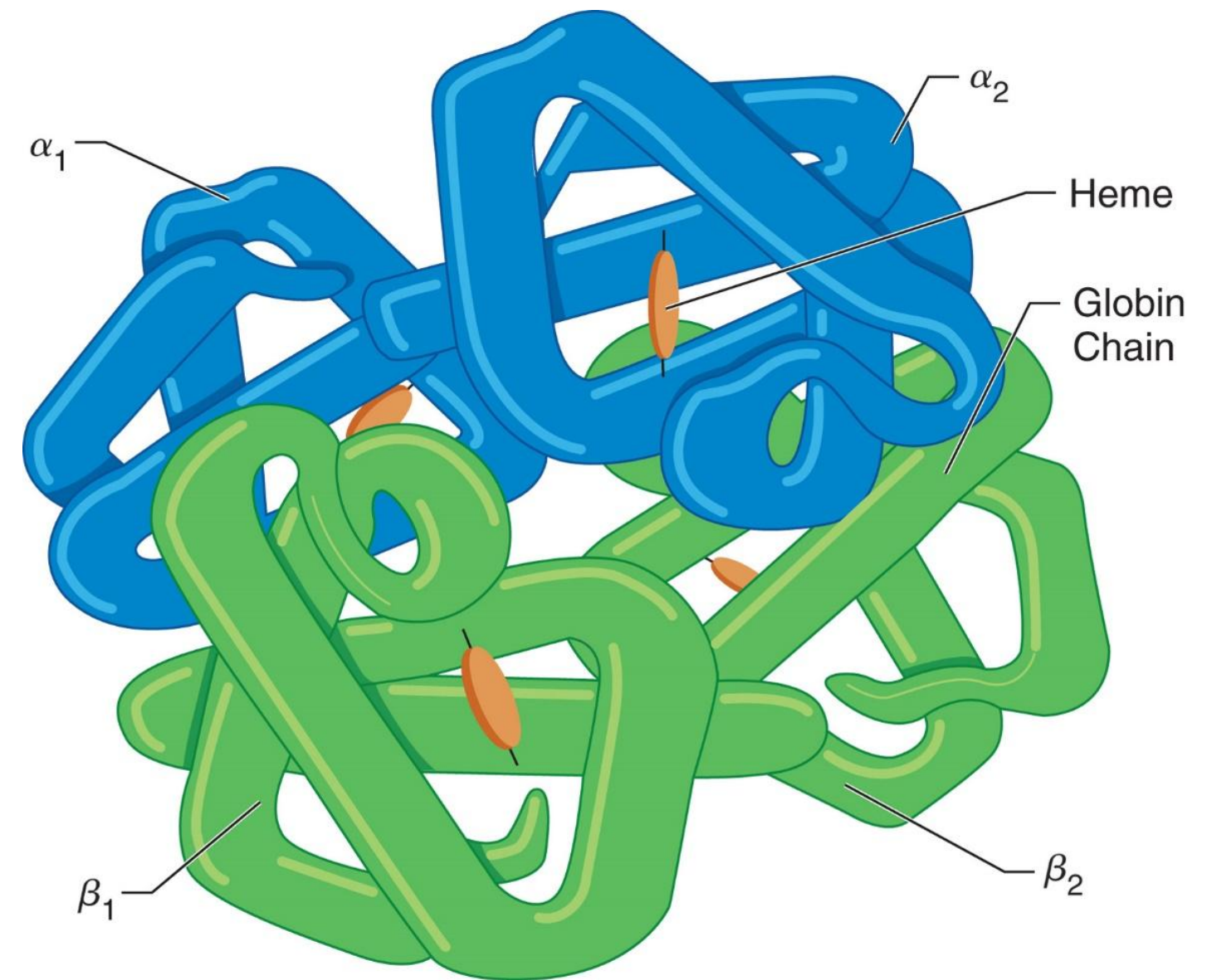
Hemoglobin

- Transports O_2 from lungs to tissues
- Facilitates CO_2 transfer from tissues to lungs
- 33% volume of RBC
 - ~90% of RBC dry weight
 - MCH = 28 – 34 pg
 - MCHC = 32 – 36 g/dL
- Total Hgb = 12 – 17 g/dL



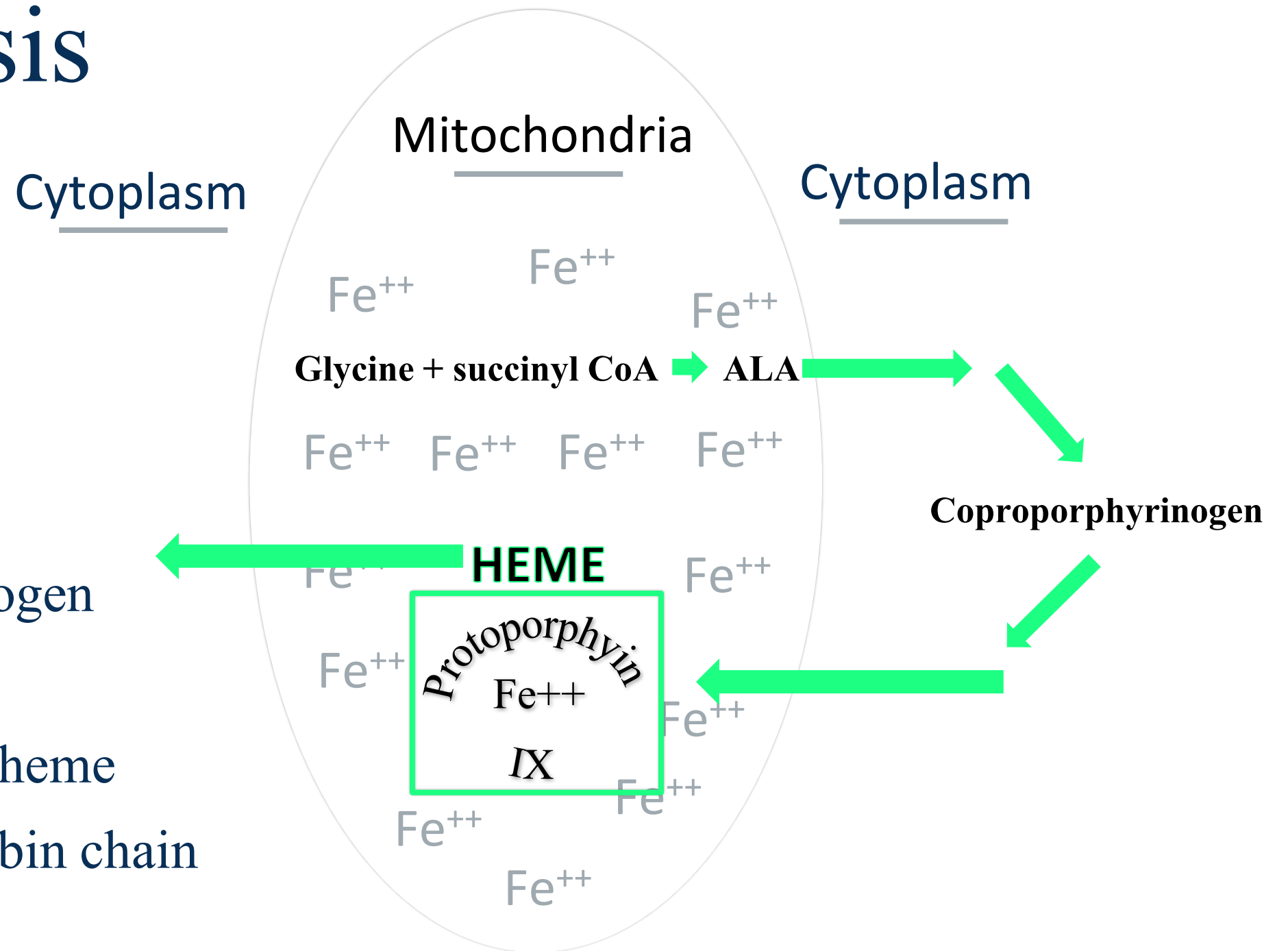
Hemoglobin Structure

- Four globular proteins
 - Two identical alpha-like
 - Alpha (α), zeta (ζ)
 - Two identical non-alpha
 - Epsilon (ϵ), beta (β), delta (δ), gamma (γ)
- Four subunits of heme
 - Iron-chelated porphyrin ring



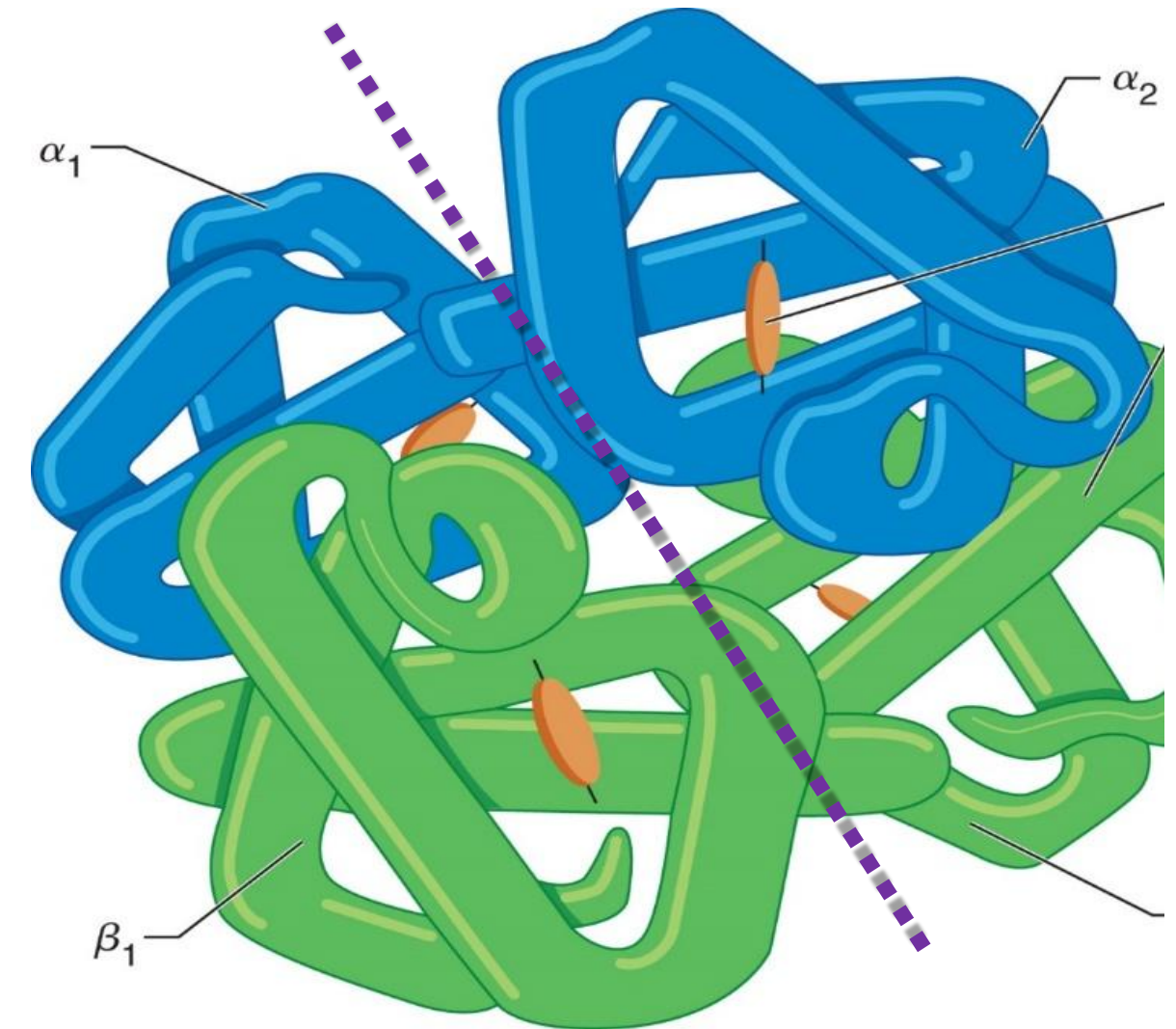
Heme Synthesis

1. Begins in mitochondria
 - Glycine + succinyl CoA $\Rightarrow$ ALA
 - [iron] limits rate of ALA production
2. Continues in cytoplasm
 - Eventual production of Coproporphyrinogen
3. Reenters and finishes in mitochondria
 - Protoporphyrin IX chelates with iron $\Rightarrow$ heme
4. Heme enters cytoplasm to bind with a globin chain



Globin Synthesis

- ζ and ϵ found only in embryonic
- After birth, α - and β -chain production predominates
 - ~97% of adult hemoglobin
- Released from polyribosomes in cytoplasm
 - Heme falls into hydrophobic pocket
- Dimers form into tetrameric hemoglobin



Chromosome 16

Chromosome 11

Globin chain

ζ

α α

ϵ

γ^G γ^A δ β

Hemoglobin Synthesis Regulation

- Normally equal production of α -, non- α -subunits, and heme
- Regulators
 - Concentration of iron
 - Concentrations of enzymes in heme production
 - Some depend on presence iron
 - Activity rate of DNA to mRNA
 - $\uparrow$ heme inactivates translation inhibitor

Hemoglobin Types

- Embryonic

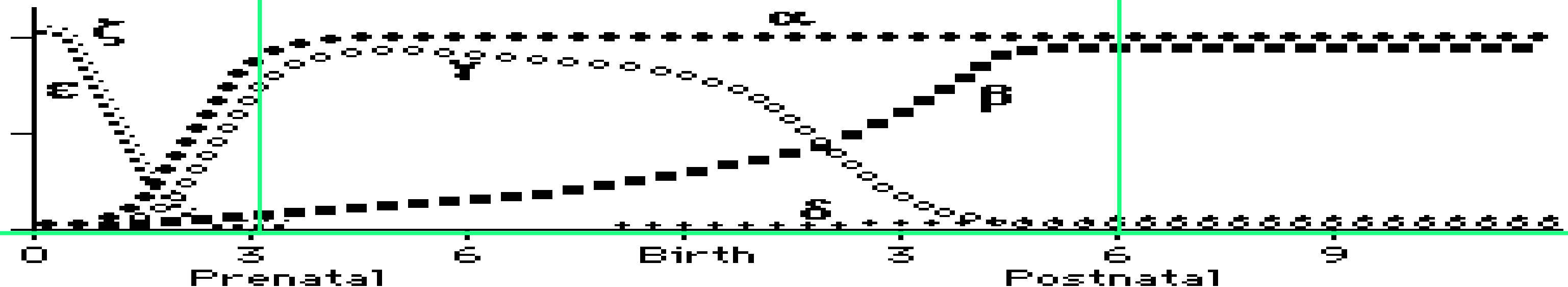
- Gower I ($\zeta_2\varepsilon_2$)
- Gower 2 ($\alpha_2\varepsilon_2$)
- Portland ($\zeta_2\gamma_2$)

- Fetal

- HbF ($\alpha_2\gamma_2$)

- Adult

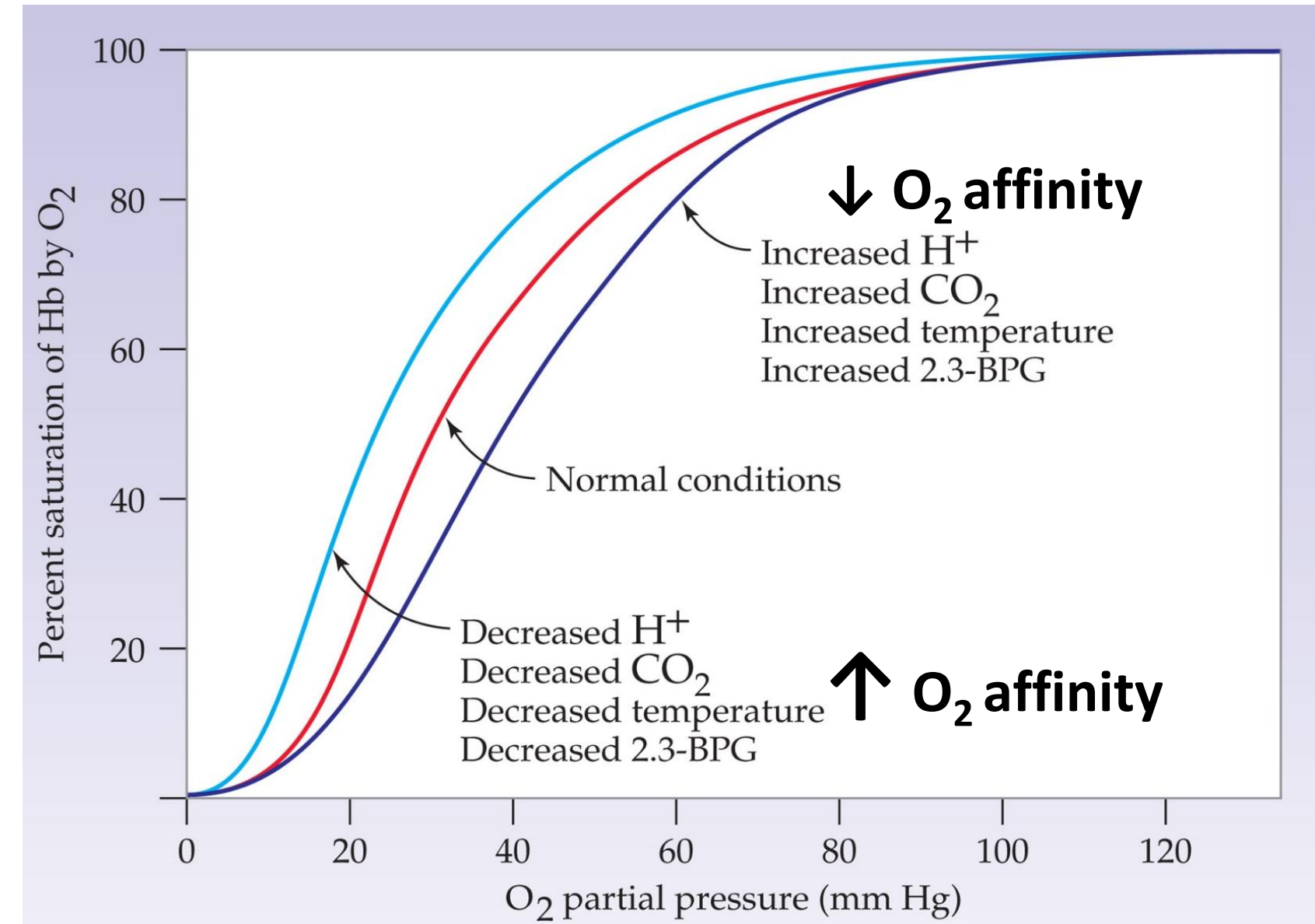
- > 95% HbA ($\alpha_2\beta_2$), includes ~ 3.5% HbA_{1c}
- < 2% HbF ($\alpha_2\gamma_2$)
- 1.5–3.5% HbA₂ ($\alpha_2\delta_2$)



Oxygen Transport

- Oxy (R)- vs Deoxy (T)- hemoglobin
- Amount of O₂ bound/released depends
 - RBC environment
 - PO₂, PCO₂, and Hgb affinity for O₂
 - P₅₀ = PO₂ when 50% Hgb saturated with O₂,
 - Occurs at ~ 26 torr = 26 mm Hg
 - ↑ P₅₀ => ↓ O₂ affinity
 - ↓ P₅₀ => ↑ O₂ affinity
 - Oxygen Dissociation Curve

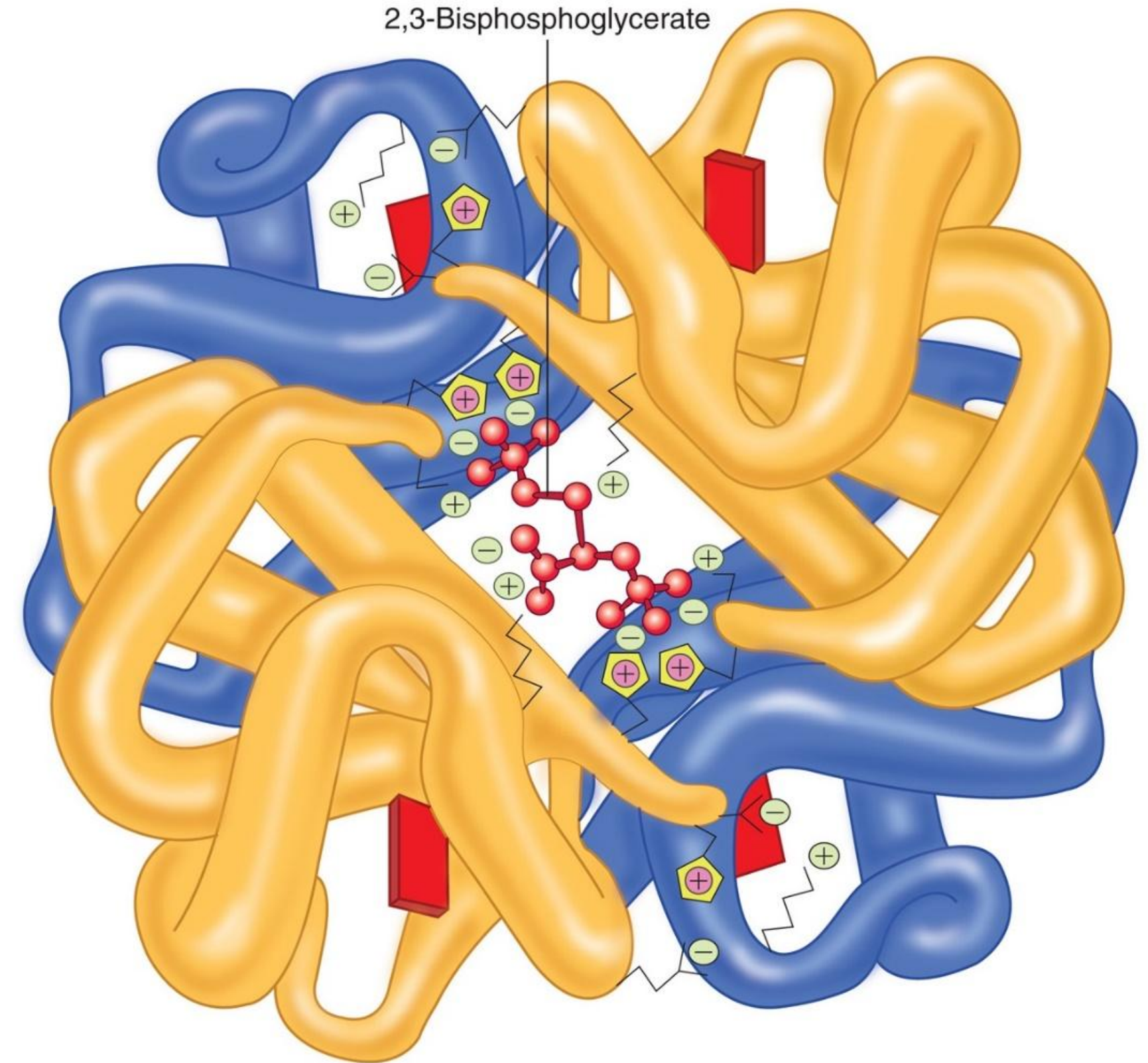
Oxygen Dissociation Curve



Allosteric Property

Hgb structure and function affected by other molecules

- 2,3-BPG (2,3-DPG)
 - Binds to T ($\downarrow$ affinity) DeoxyHgb in 1:1
 - 3rd O₂ expels 2,3-BPG
 - $\uparrow$ O₂ affinity



CO₂ Transport

- CO₂ carried to lungs via
 - ~ 70% Formation of carbonic acid
 - ~ 23% Bound to Hgb
 - Binds deoxyhgb, expired in lungs
 - ~ 7% Dissolution into plasma

Hemoglobin Destruction

- Extravascular
 - Most efficient and recycles RBC components
 - Iron
 - Stored as ferritin or hemosiderin in macrophages
 - Transported to BM via transferrin (80% of pool)
 - Globin
 - Broken down into amino acids
 - Heme cleaved to CO (lungs) + biliverdin (bilirubin)

Hemoglobin Destruction

- Intravascular
 - Dissociates into dimers
 - Quickly binds to haptoglobin (liver)
 - Acute hemolysis
 - Haptoglobin depleted
 - Globin dimers filtered by kidney
 - Reabsorbed in proximal tubules
 - » Hemosiderinuria

Acquired Nonfunctional Hgbs

- Patients present with hypoxia and/or cyanosis
 - Methemoglobin (no O₂ affinity)
 - Hgb iron in ferric state (Fe⁺⁺⁺) = MetHgb
 - Blood appears chocolate brown
 - Sulfhemoglobin
 - Sulfur atom binds to heme for life of RBC, no O₂ affinity
 - Carboxyhemoglobin
 - Hgb 200x ↑ CO affinity compared to O₂
 - Cherry red blood/skin